

Diffuse Parenchymal Lung Diseases (Restrictive Lung Disease): An Approach to a Case of Dyspnoea

Dr. Essam Kotb, Associate Professor, Internal Medicine | Source: 9-

Restrictive_lung_disease_Updated.pdf (35 slides) · Refs: Step-Up 5th ed.; Harrison 20th ed.; UpToDate; Osmosis; AMBOSS

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from earlier respiratory lectures:** COPD and asthma were **obstructive** (FEV1/FVC < 70%). Here the pattern flips: **restrictive** = small, stiff lungs with a **normal or high FEV1/FVC** and **low volumes**. The **spirometry algorithm** shown in the COPD lecture is reused here, now taking the "FVC reduced → restriction" branch. Links back: **asbestos** (lung cancer, mesothelioma, pleural plaques), **silicosis → TB**, **sarcoidosis → hypercalcaemia** (compare squamous-cell PTHrP), and ILD → **Group 3 pulmonary hypertension** (Pulmonary Vascular lecture).

Restrictive lung disease: spot the pattern, then the disease

PFT: FEV1/FVC normal or ↑ · ALL volumes ↓ (TLC, FVC, FRC, RV) · DLCO ↓ if parenchymal (normal if extrinsic [added])

Intrinsic = stiff lung (fibrosis, granulomas, alveolar filling) · Extrinsic = pleura, chest-wall deformity, obesity/ascites/pregnancy, weak muscles

IPF

> 50 y, slow, no fever
Velcro crackles, clubbing
HRCT: basal subpleural
HONEYCOMBING (UIP)

Pirfenidone / nintedanib
transplant · NO steroids

COP

Days-weeks, flu-like,
mimics CAP
Bilateral PATCHY opacities,
normal lung volumes

Steroid-responsive
50% idiopathic

SARCOIDOSIS

Young (< 40), African-
American women
NONCASEATING granulomas
Bilateral hilar nodes, ↑ Ca²⁺
↑ ACE, erythema nodosum

Steroids ± DMARD

HYPERSENSITIVITY PNEUMONITIS

Organic antigen (birds,
mouldy hay, hot tubs)
Acute: fever 4–8 h after
exposure; ground glass

Avoid agent ± steroids

Pneumoconioses (inorganic dust → macrophages → fibrosis)

ASBESTOSIS

LOWER lobes · > 15–20 y latency
Pleural plaques (calcified)
↑ bronchogenic cancer (smoking
synergistic) + mesothelioma
No specific treatment

SILICOSIS

UPPER lobes · nodular
"Eggshell" calcified hilar nodes
↑ TB (impaired phagolysosome)
→ annual TST
Mining, sandblasting, glass

BYSSINOSIS / BERYLLIOSIS

Byssinosis: raw cotton dust
(textile workers)
Berylliosis: sarcoid-like
granulomas; steroids +
avoid beryllium

Added diagram: restrictive lung disease at a glance

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Define restrictive disease; classify DPLDs (idiopathic, exposure-related, systemic)	Intrinsic vs extrinsic; DPLD tree
Knowledge	★ Mechanisms: injury → fibroblasts → matrix → fibrosis; particle phagocytosis (pneumoconioses); non-caseating granulomas (sarcoidosis)	Pathogenesis
Knowledge	★ Signs, extrapulmonary features, complications (pulmonary HTN, ↑ cancer and infection)	Velcro crackles; sarcoid organs; silica → TB; asbestos → cancer

Domain	Objective	Likely tested as
Knowledge	★ PFT, imaging and treatment findings	Restrictive PFT + ↓ DLCO; honeycombing; antifibrotics
Skills	Differentiate intrinsic vs extrinsic restriction; plan diagnosis and treatment (sarcoid, drug-induced)	Case vignettes
Values	Professionalism with complex respiratory patients	

Flow: Overview (restrictive disease, intrinsic vs extrinsic) → DPLDs (definition, pathophysiology, features, CXR/CT/PFT, classification) → IPF → COP → AIP/NSIP → Radiation pneumonitis → Treatment (IPF vs non-IPF) → Sarcoidosis → Hypersensitivity pneumonitis → Pneumoconioses (asbestosis, silicosis, byssinosis) → Cases.

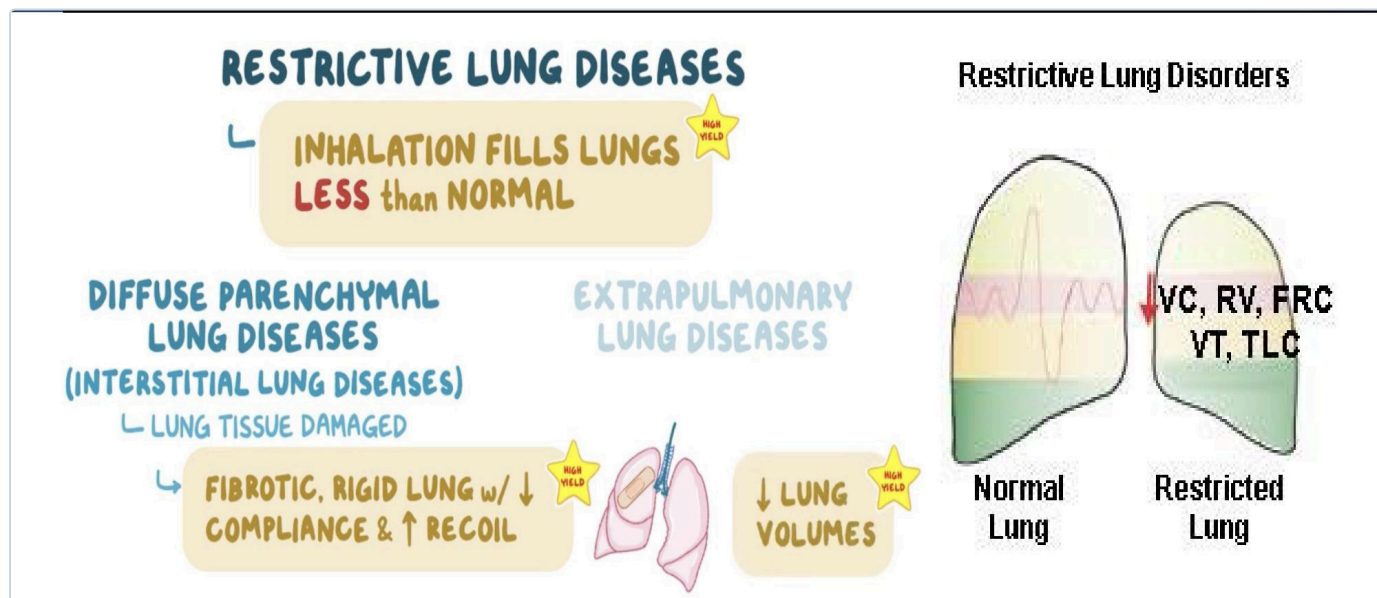
2. Main Content (Lecture Order)

2.1 Overview: Restrictive Lung Disease (Slides 5-6)

★ **Restrictive** = **inhalation fills the lungs less than normal** → **↓ lung volumes**. Two families:

- ★ **Diffuse parenchymal (interstitial) lung disease:** lung tissue damaged → **fibrotic, rigid lung** with **↓ compliance, ↑ recoil**
- ★ **Extrapulmonary:** lung normal, but expansion is limited

★ Intrinsic (parenchymal)	★ Extrinsic (extrapulmonary mechanics)
ILD / DPLD · sarcoidosis · pneumoconioses · ARDS · IPF · hypersensitivity pneumonitis (extrinsic allergic alveolitis) · granulomatosis with polyangiitis · radiation lung injury · drug-induced ILD · alveolar filling (pneumonia, oedema, haemorrhage)	Pleural disease (chronic effusion, pneumothorax) · thoracic deformity / mechanical limitation (kyphoscoliosis, ankylosing spondylitis, pregnancy, ascites) · respiratory-muscle weakness (phrenic palsy, myasthenia gravis)

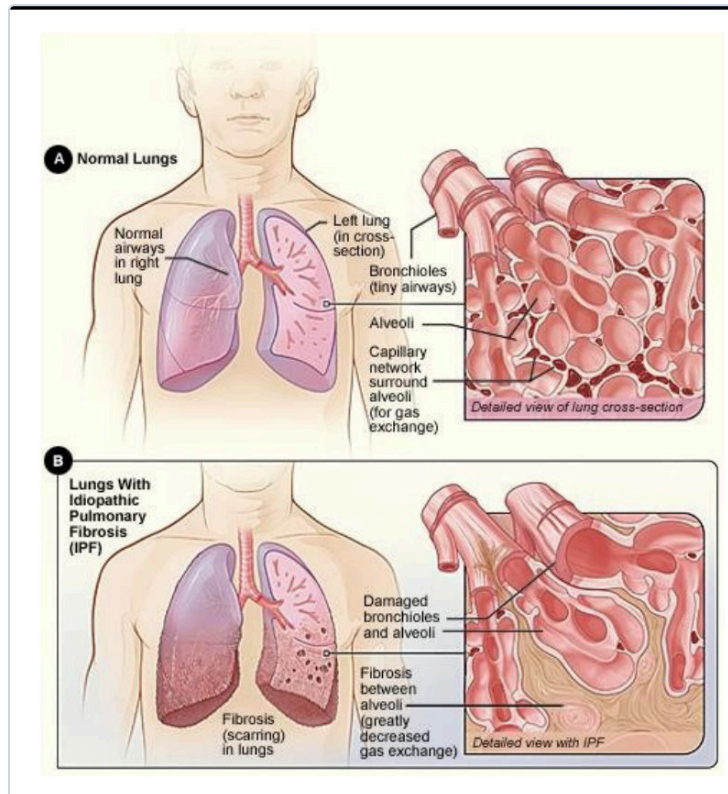


Slide 5: Restrictive lung diseases: smaller lung volumes (↓ VC, RV, FRC, VT, TLC)

2.2 Diffuse Parenchymal Lung Diseases: Definition & Mechanism (Slides 7-8)

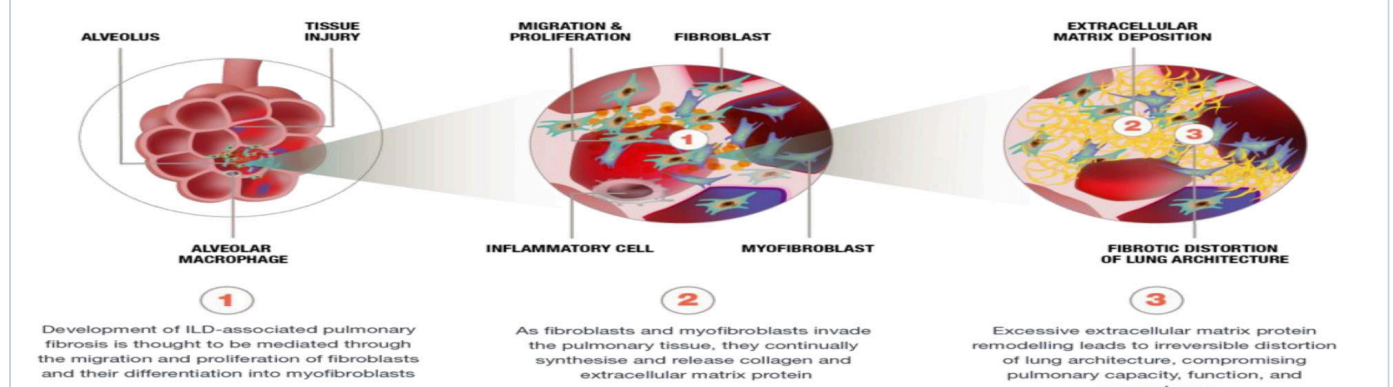
★ **DPLD** (formerly **interstitial lung disease, ILD**): a **diverse group of rare, highly morbid disorders** with a **progressive inflammatory process** that leads to **widespread scarring (fibrosis)**.

★ **Shared pathophysiology: repeated tissue injury** in the parenchyma + **aberrant wound healing** → **collagenous fibrosis** → **remodelling of the interstitium**. *Sequence on the figure: alveolar injury → inflammatory cells → fibroblast proliferation → extracellular matrix deposition.*



Slide 7: Normal lung vs IPF (fibrosis, damaged alveoli)

- Repeated cycles of tissue injury in the lung parenchyma with aberrant wound healing → collagenous fibrosis → remodeling of the pulmonary interstitium



Slide 8: ILD pathogenesis: injury → inflammation → fibroblasts → matrix

2.3 Clinical Features (Slide 9)

- ★ **Progressive dyspnoea** (exertion first, then at rest)
- ★ **Persistent dry cough**
- ★ **Bibasilar inspiratory "Velcro" (fine) crackles**

- **Fatigue**; ★ **clubbing**
- **Fever** (in **hypersensitivity pneumonitis** and **sarcoidosis**)
- Clues to **connective tissue disease, sarcoidosis, vasculitis**
- **Advanced**: clubbing (chronic hypoxia), **cyanosis**, loud inspiratory wheeze
- *Figure*: complication = **pulmonary arterial hypertension**

Interstitial Lung Diseases

Diffuse Parenchymal Lung Diseases

INFLAMMATION AND/OR FIBROSIS IN LUNG TISSUE.

RESTRICTIVE —

- ✓ REDUCED TLC, FVC, FEV1

SIGNS & SYMPTOMS

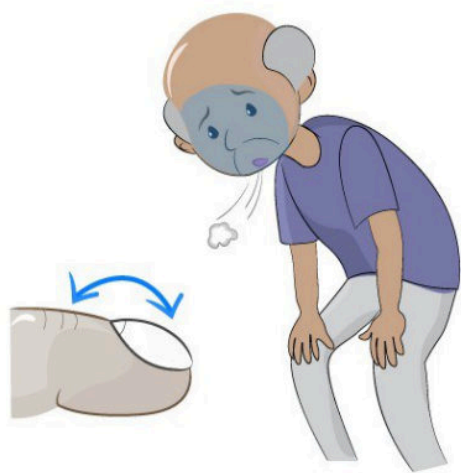
- ✓ PROGRESSIVE DYSPNEA
ON EXERTION -> AT REST
- ✓ DRY COUGH
- ✓ DIGIT CLUBBING
- ✓ LUNG OPACITIES ON CXR

COMPLICATIONS

- ✓ PULM. ARTERIAL HYPERTENSION

DIAGNOSIS

- ✓ DIFFICULT DUE TO NON-SPECIFIC,
OVERLAPPING SIGNS AND SYMPTOMS.



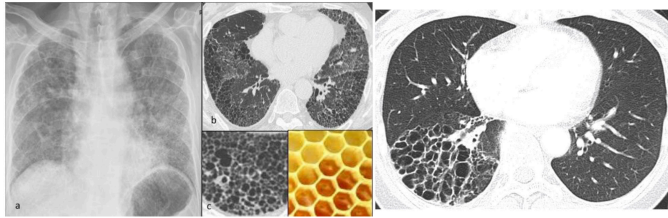
The illustration shows a person in a purple shirt and grey pants, bent over with their hands on their knees, appearing to be coughing or in distress. A small cloud of white smoke or cough is coming from their mouth. To the left of the person is a close-up of a hand showing digital clubbing, with a blue double-headed arrow indicating the swelling at the tip of the finger.

Slide 9: ILD signs and symptoms

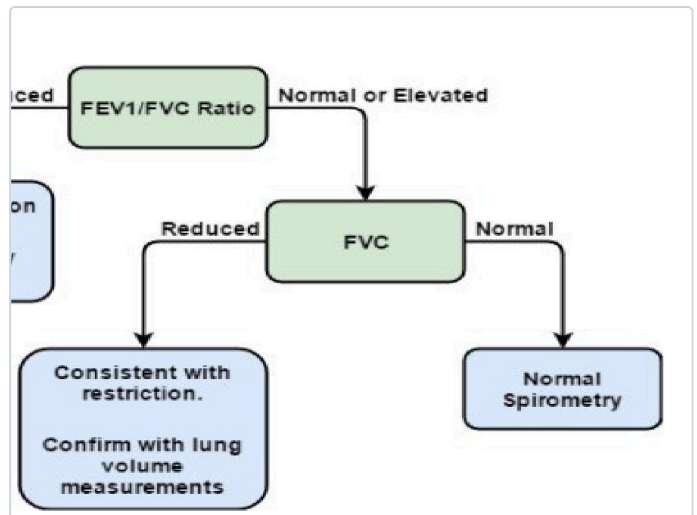
2.4 Diagnostics (Slides 10-12)

Test	★ Findings
CXR	Diffuse opacification: reticular, reticulonodular, ground glass, honeycombing
★ Chest CT (HRCT)	Reticulation, ground glass, ★ honeycombing
★★ PFTs	★ Restrictive pattern: FEV1/FVC normal or ↑ · ★ ↓ ALL lung volumes · ★ ↓ DLCO

Diagnostics:
Chest CT



Slide 11: CXR and HRCT honeycombing (with the honeycomb analogy)



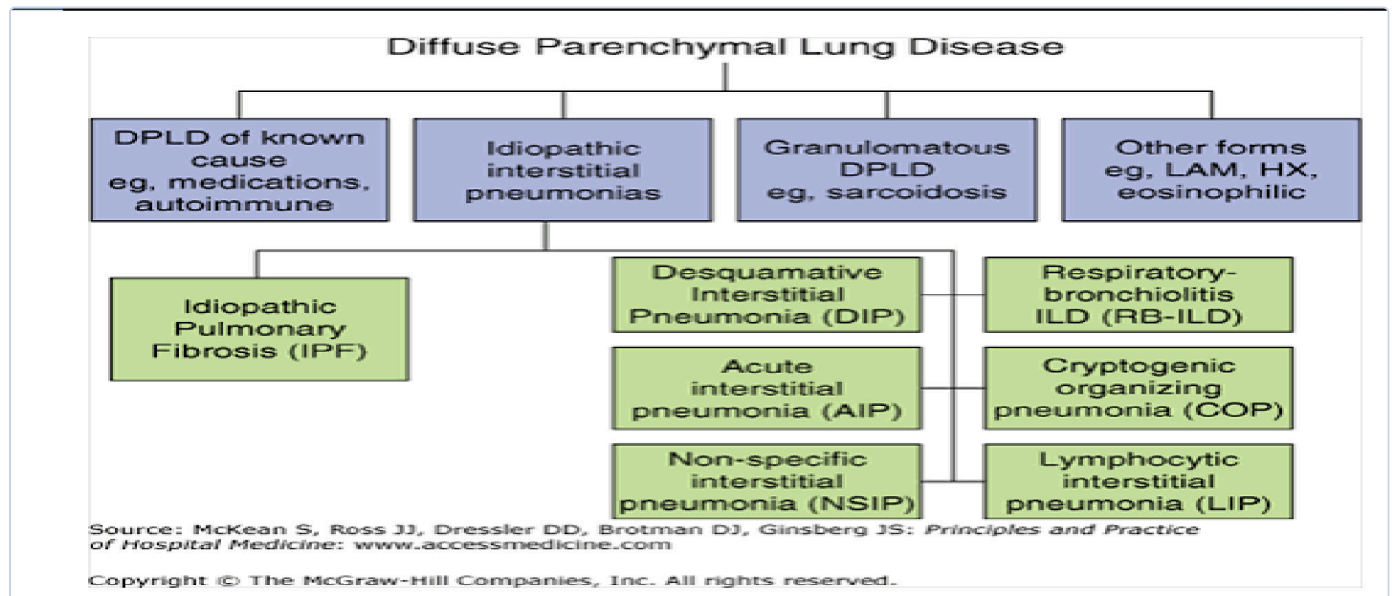
Slide 12: Spirometry algorithm: normal/↑ ratio + ↓ FVC = restriction

[added: DLCO separates intrinsic (↓) from extrinsic restriction (usually normal), since the gas-exchange membrane is healthy in chest-wall or muscle disease.]

2.5 Classification (Slide 13)

★ DPLD →

1. **Known cause: medications, autoimmune** (connective tissue disease)
2. ★ **Idiopathic interstitial pneumonias:** ★ **IPF**, desquamative interstitial pneumonia (**DIP**), respiratory-bronchiolitis ILD (**RB-ILD**), acute interstitial pneumonia (**AIP**), cryptogenic organising pneumonia (**COP**), non-specific interstitial pneumonia (**NSIP**), lymphocytic interstitial pneumonia (**LIP**)
3. ★ **Granulomatous:** e.g., **sarcoidosis**
4. **Other:** e.g., **LAM, histiocytosis X, eosinophilic**



Slide 13: DPLD classification

🔗 **Recap: DPLD basics.** Restrictive = low volumes, normal/high FEV1/FVC, low DLCO (if parenchymal). Intrinsic (fibrosis, granulomas, alveolar filling) vs extrinsic (pleura, chest wall, muscles). Injury + abnormal healing → fibrosis. Dyspnoea, dry cough, Velcro crackles, clubbing. HRCT honeycombing.

🔗 **Link:** The most common and most important idiopathic type comes next.

📌 **Up next: IPF and other idiopathic types.**

2.6 Idiopathic Pulmonary Fibrosis (Slide 14)

★ **The most common ILD: irreversible fibrosis**, impaired lung function.

	Details
★ Who	Middle-aged or older (> 50 y); chronic dry cough, slowly worsening dyspnoea over months
Course	Not acutely ill, no fever; very slow progression
★ Exam	Bibasilar Velcro crackles; restrictive PFT (↓ FVC, TLC, FRC), ★ ↓ DLCO
★ Diagnosis	CXR diffuse infiltrates (non-specific); ★ biopsy: usual interstitial pneumonia (UIP); ★ HRCT: bibasilar reticular patchiness + honeycombing + traction bronchiectasis in subpleural areas
Prognosis	★ Poor despite immunosuppression; transplant improves survival and quality of life

impaired pulmonary function:

	SYMPTOMS/EXAM	DIAGNOSIS	TREATMENT
Idiopathic pulmonary fibrosis (IPF)—the most common ILD	Affects middle-age or older patients (>50 years); presents with chronic, nonproductive cough and gradually worsening dyspnea over many months Not acutely ill and do not present with fever; very slow progression Exam reveals bibasilar inspiratory crackles (Velcro-like), restrictive lung disease (↓ FVC, TLC, FRC), and ↓ gas exchange (DLCO)	CXR with diffuse infiltrates (nonspecific) Lung biopsy reveals usual interstitial pneumonia; HRCT shows bibasilar, reticular patchiness with honeycombing and bronchiectasis in subpleural areas	Has a poor prognosis despite immunosuppressive therapy; lung transplantation can improve survival and quality of life

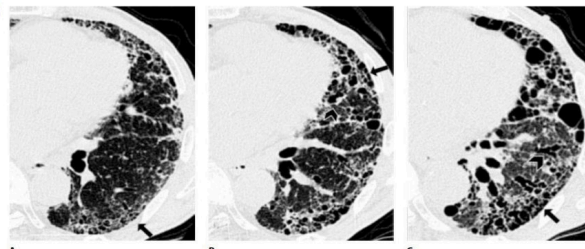


FIGURE 4-9. Idiopathic pulmonary fibrosis (IPF). HRCT findings of a 78-year-old man with IPF who died 20 months after initial diagnosis. Lower left lobe shows fibrosing changes at baseline (A), 6 months (B), and 12 months (C). Changes include subpleural predominant interstitial fibrosis, traction bronchiectasis (arrowheads), and honeycombing (arrows). The overall HRCT fibrosis score at the baseline, 6 months, and 12 months were 151.3, 162.5, and 185.8, respectively. (Reproduced with permission from USMLE-Rx.com.)

Slide 14: IPF HRCT: progressive subpleural honeycombing over 12 months

2.7 Cryptogenic Organising Pneumonia (Slide 15)

Rare ILD: inflammation of **bronchioles, alveolar ducts and alveolar walls**.

- ★ **Acutely ill over days to weeks;** ★ **mimics community-acquired pneumonia** (flu-like: fever, fatigue, dry cough); exertional dyspnoea, weight loss
- ★ **50% idiopathic**, 50% associated with toxic fumes, drugs, infection, connective tissue disease, immunodeficiency, radiation
- ★ **CXR/CT: bilateral patchy opacities with normal lung volumes**
- Biopsy shows organising pneumonia (not diagnostic on its own)
- ★ **Responds to corticosteroids**

alveolar ducts, and alveolar walls

Cryptogenic
organizing
pneumonia

Acutely ill over days to weeks, and mimics community-acquired pneumonia symptoms and may have initial **flulike** symptoms (eg, fever, fatigue, nonproductive cough); dyspnea on exertion and weight loss are also common

50% are idiopathic, 50% are associated with toxic fume inhalation, certain drug exposures, infections, connective tissue disorders, immunodeficiencies, radiation

CXR and CT show bilateral **patchy** opacities with normal lung volumes
Lung biopsy reveals organizing pneumonia but is not diagnostic

Responsive to corticosteroids



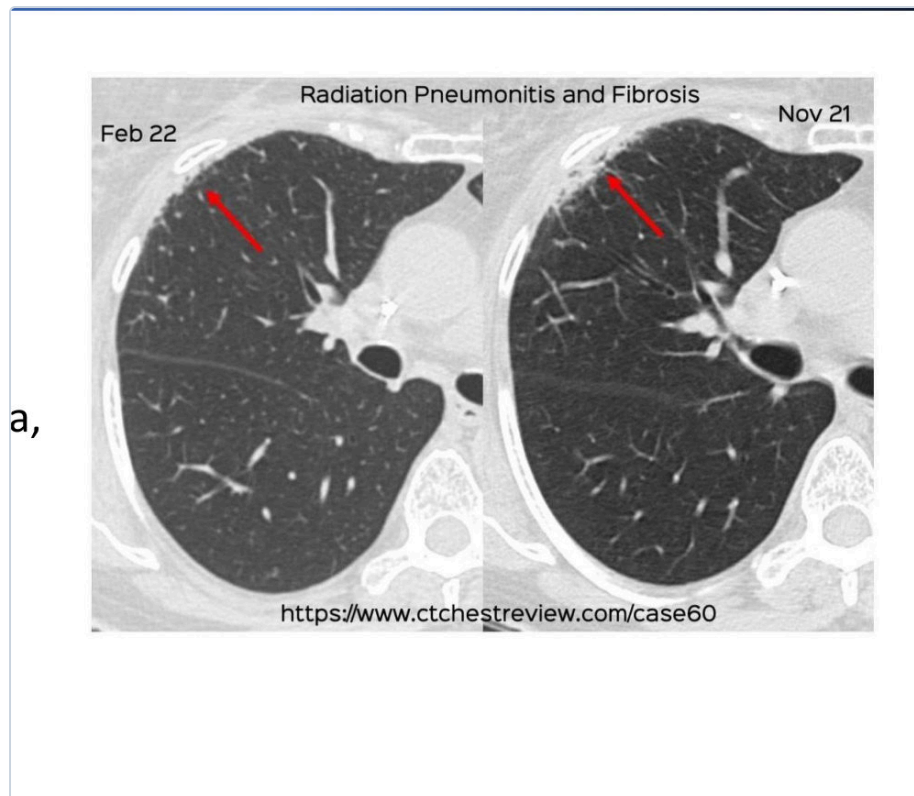
Slide 15: COP: bilateral patchy opacities

2.8 Other Idiopathic Types (Slide 16)

- ★ **Acute interstitial pneumonia (AIP, Hamman-Rich syndrome):** severe, acute, can progress rapidly to **respiratory failure**
- ★ **Non-specific interstitial pneumonia (NSIP):** associated with **connective tissue disease (e.g., systemic sclerosis), HIV, hypersensitivity pneumonitis**
- Distinguished **histologically**

2.9 Radiation Pneumonitis (Slide 17)

- Occurs in ★ **5-15%** of patients given thoracic radiotherapy
- ★ **Acute: 1-6 months** after; **chronic: 1-2 years** after
- Alveolar thickening and fibrosis**
- Low-grade fever, cough, chest fullness, dyspnoea, pleuritic pain, haemoptysis
- ★ **CT is the best study;** ★ **corticosteroids are the treatment of choice**



Slide 17: Radiation pneumonitis progressing to fibrosis on CT

2.10 Treatment (Slide 18)

★★ IPF:

- **Antifibrotics** (may ↓ mortality and acute exacerbations; slow FVC decline):
 - ★ **Pirfenidone**: inhibits **TGF- β -stimulated collagen synthesis**
 - ★ **Nintedanib**: inhibits **tyrosine kinases** of fibrogenic growth factors
- ★ **Refer for lung transplant at diagnosis**
- ⚠ ★ **Do NOT start immunosuppression in IPF**: it may **worsen outcomes**

Non-IPF ILD:

- **Treat the underlying condition**; antibiotics if bacterial interstitial pneumonia is suspected
- **Consider corticosteroids and/or immunomodulators**
- **Lung transplant**

🔄 **Recap: Idiopathic ILDs.** IPF (> 50 y, slow, no fever, UIP, basal subpleural honeycombing, poor prognosis) → pirfenidone/nintedanib + transplant referral, **no steroids**. COP (acute, mimics pneumonia, patchy opacities with normal volumes) → **steroids work**. AIP = rapid failure; NSIP = CTD/HIV. Radiation pneumonitis 1-6 months after radiotherapy → steroids.

🔗 **Link:** Granulomatous ILD behaves differently: young patients, many organs, often remits.

👉 **Up next: Sarcoidosis.**

2.11 Sarcoidosis (Slides 19-23)

★★ General characteristics:

- ★ **Chronic systemic granulomatous disease** with ★ **non-caseating granulomas**, often **multi-organ**

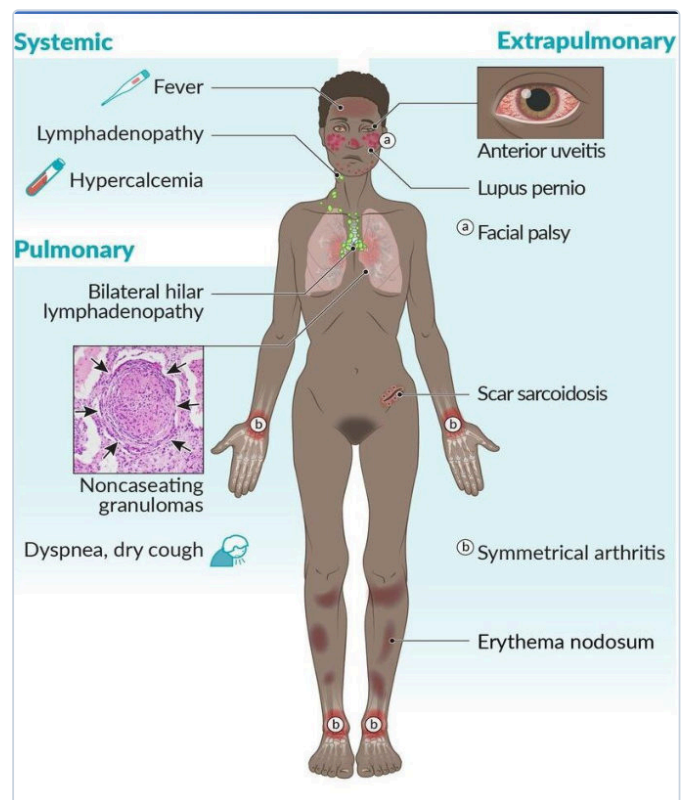
- **Cause unknown:** genetic, environmental (e.g., **beryllium**), infections
- ★ Most often **African-American women, < 40 years**
- ★ **Good prognosis in most:** ~ $\frac{2}{3}$ **spontaneous remission**, ~ $\frac{1}{3}$ **chronic progressive**

★ **Presentation (Slides 20-21):**

Area	Features
General	Often asymptomatic early; fever, malaise, anorexia, weight loss, ★ lymphadenopathy
Pulmonary	Dyspnoea, cough, chest pain, rales; ★ bilateral hilar lymphadenopathy
★ Hypercalcaemia	Granuloma macrophages make 1,25-(OH)₂ vitamin D (calcitriol)
Skin (~30%)	★ Erythema nodosum (commonest, non-specific) · ★ lupus pernio (pathognomonic : violaceous plaques on nose, cheeks, chin, ears)
Joints / eyes	Polyarthralgia/arthritis · ★ anterior uveitis
Liver / spleen / heart	Liver disease, splenomegaly, cardiomyopathy
★ Neurosarcoidosis	★ Facial nerve palsy (uni/bilateral) , diabetes insipidus , hypopituitarism, meningitis, neuropathy, myopathy
Kidneys	Interstitial nephritis, nephrocalcinosis, nephrolithiasis



Slide 20: Lupus pernio (pathognomonic)



Slide 21: Sarcoidosis systemic and extrapulmonary features

★★ **Diagnosis (Slide 22): a diagnosis of exclusion**

- ★ **Biopsy: non-caseating granulomas**
- **CXR, HRCT**
- **LFTs, ↑ calcium, BUN/creatinine; urinalysis (hypercalciuria)**
- ★ **↑ Angiotensin-converting enzyme (ACE)**
- **Eye exam, ECG** (cardiac sarcoid)
- **Spirometry:** usually **restrictive** (obstructive or mixed possible); ★ **↓ DLCO**

★★ CXR staging (Table 4.16):

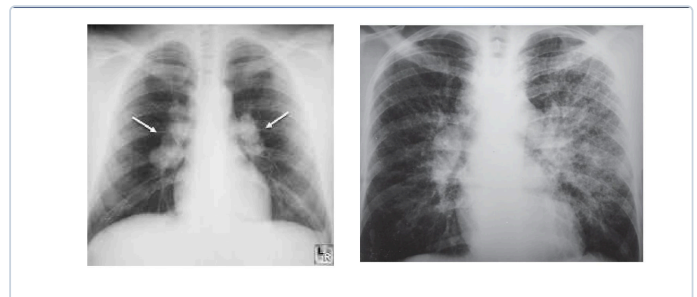
Stage	CXR	Spontaneous resolution
0	Normal	n/a
1	★ Bilateral hilar lymphadenopathy	★ 60-80%
2	Hilar nodes + reticulonodular opacities	50-60%
3	Reticulonodular opacities without nodes	< 30%
4	Fibrosis	n/a

★ **Treatment: corticosteroids**; consider adding a **DMARD** [added: e.g., methotrexate as a steroid-sparing agent]

TABLE 4.16. Stages of Sarcoidosis

	CXR FINDINGS	SPONTANEOUS RESOLUTION WITHOUT TREATMENT
Stage 0	Normal	—
Stage 1	Hilar lymphadenopathy (see Figure 4.11)	60%-80%
Stage 2	Hilar lymphadenopathy and reticulonodular opacities	50%-60%
Stage 3	Reticulonodular opacities (no lymphadenopathy)	<30%
Stage 4	Fibrotic changes	—

Slide 22: Sarcoidosis stages



Slide 23: CXR: bilateral hilar lymphadenopathy

📌 **Recap: Sarcoidosis.** Non-caseating granulomas; young African-American women. Bilateral hilar nodes, hypercalcaemia (calcitriol), ↑ ACE, erythema nodosum, lupus pernio, uveitis, facial palsy. Diagnosis of exclusion with biopsy. Staging 0–4 by CXR (stage 1 remits 60–80%). Steroids ± DMARD.

🔗 **Link:** Some ILDs are caused by what you breathe at work or home: organic antigens (hypersensitivity) and inorganic dust (pneumoconioses).

📌 **Up next: Hypersensitivity pneumonitis.**

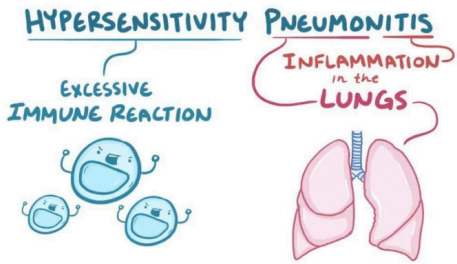
2.12 Hypersensitivity Pneumonitis (Slide 24)

★ **Allergic reaction to inhaled ORGANIC agents** (= extrinsic allergic alveolitis).

Phase	Features
★ Acute	Fever, chills, cough, dyspnoea 4-8 hours after exposure ; inspiratory crackles; ★ ground-glass opacities on CT
Chronic	Gradual cough, dyspnoea, malaise, weight loss; diffuse fibrosis on biopsy
★ Treatment	★ Avoid the agent (mould, dust, bird droppings , aerosolised water/ hot tubs , pesticides); corticosteroids if acute and severe

★ **Causes ("Quick hit"):** **farmer's lung** (mouldy hay) · ★ **bird-breeder's lung** (avian droppings) · air-conditioner lung · **bagassosis** (mouldy sugar cane) · mushroom-worker's lung (compost)

CATEGORY	PRESENTATION
OCCUPATIONAL/EXPOSURE	
Hypersensitivity pneumonitis (organic inhalations)	Caused by an allergic reaction to inhaled organic agents Acute symptoms: Fever, chills, cough, dyspnea 4-8 hours after exposure; inspiratory crackles; ground-glass opacities on CT Chronic symptoms: Gradual cough and dyspnea, malaise, weight loss; diffuse fibrosis on biopsy Treatment: Avoid common inciting agents such as mold, dust, bird droppings, aerosolized/humidified water (hot tubs), and pesticides; give corticosteroids if acute and severe



Quick HIT ⚡

Some causes of hypersensitivity pneumonitis:

- Farmer's lung (moldy hay)
- Bird-breeder's lung (avian droppings)
- Air-conditioner lung
- Bagassosis (moldy sugar cane)
- Mushroom worker's lung (compost)

Slide 24: Hypersensitivity pneumonitis: presentation and causes

2.13 Pneumoconioses (Slides 25-30)

★ **Definition:** restrictive ILDs from **inhaled (mainly inorganic) dust**, often in **miners and agricultural workers**. ★ **Mechanism:** dust inhaled → **phagocytosis by alveolar macrophages** → **macrophage destruction + inflammation** → **scarring, granulomas**.

Disease	Agent	Exposed workers
★ Asbestosis	Asbestos fibres	Asbestos mining/milling, brake linings, insulation , ★ shipbuilding , firefighters, plumbers
★ Silicosis	Crystalline silica	Mining, quarries, pottery, sandblasting, glass manufacturing, construction
Byssinosis	Raw cotton dust (less often flax, jute, hemp, sisal)	Textile workers

★★ Asbestosis (Slide 26):

- **Diffuse interstitial fibrosis**, ★ **lower lobes**
- **Insidious:** ★ **> 15-20 years after exposure**
- ★ **↑ bronchogenic carcinoma (smoking is synergistic)** and ★ **malignant mesothelioma**
- Diagnosis: clinical + exposure history
- CXR: bilateral linear opacities; ★ **pleural plaques** (lower zones)
- **No specific treatment**
- (Slide 29): follow up for **bronchogenic adenocarcinoma** or **mesothelioma**, especially if **exposed > 30 years**; ★ **adenocarcinoma is the commonest lung cancer with asbestos**

★★ Silicosis (Slide 27):

- **Localised, nodular peribronchial fibrosis**, ★ **upper lobes**
- **Acute** (massive exposure → rapid death) or **chronic** (≥ 15 years)
- ★ **↑ risk of TB:** silica **impairs phagolysosome formation**, so **M. tuberculosis survives in macrophages** → ★ **annual tuberculin skin test** (Slide 30)
- Exertional dyspnoea; productive cough; restrictive PFT
- Treatment: **supportive + remove from exposure**
- **Quick hit:** ★ **asbestosis → pleural plaques**; ★ **silicosis → "eggshell" calcification** (of hilar nodes)

★ **Imaging summary (Slide 28):**

Finding	Disease
Bilateral hilar adenopathy	Sarcoidosis
Upper-lobe calcified nodules, eggshell calcification	Silicosis
Honeycomb parenchyma	IPF
Pleural effusions	Connective tissue disease
Linear pleural plaques	Asbestosis

Treatment (Slides 29-30): treat the cause; **avoid the occupational exposure**; ★ **quit smoking**.
Berylliosis → **steroids + avoid beryllium**. Hypersensitivity pneumonitis → **remove the agent**,
steroids if severe.

DIAGNOSIS

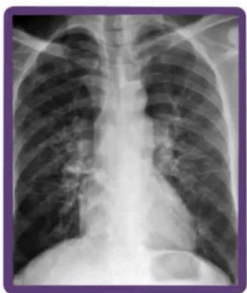
* **CHEST X-RAY**

- NON-SPECIFIC INTERLACING WEB-LIKE RETICULAR INFILTRATES
- SPHERICAL NODULAR INFILTRATES

} **RETICULONODULAR PATTERN**

* **high resolution computed tomography → HRCT SCAN**

- BILATERAL HILAR ADENOPATHY → **SARCOIDOSIS**
- UPPER LOBE CALCIFIED NODULES, called "EGGSHELL CALCIFICATIONS" → **SILICOSIS**
- LUNG PARENCHYMA → HONEYCOMB → **IDIOPATHIC PULMONARY FIBROSIS**
- PLEURAL EFFUSIONS → **CONNECTIVE TISSUE DISEASES**
- LINEAR PLEURAL PLAQUE → **ASBESTOSIS**



Slide 28: Pneumoconiosis imaging patterns

TREATMENT

- * DEPENDS on UNDERLYING CAUSE
- * OCCUPATIONAL LUNG DISEASE → AVOID OCCUPATIONAL ENVIRONMENT
- * QUIT SMOKING

ASBESTOSIS

- * FOLLOWED up for BRONCHOGENIC ADENOCARCINOMA or MESOTHELIOMA
- especially if EXPOSED > 30 YRS
- BRONCHOGENIC ADENOCARCINOMA is most COMMON TYPE of LUNG CANCER associated w/ ASBESTOS



Slide 29: Asbestosis follow-up for cancer

TREATMENT

SILICOSIS

- * SUSCEPTIBLE to TUBERCULOSIS
- SILICA IMPAIRS FORMATION of PHAGOLYSOSOME
- MYCOBACTERIUM TUBERCULOSIS SURVIVES w/in CELL
- * SCREENED ANNUALLY w/ TUBERCULIN SKIN TEST

BERYLLIOSIS

- * CORTICOSTEROIDS
- * AVOIDING EXPOSURE to BERYLLIUM

HYPERSENSITIVITY PNEUMONITIS

- * REMOVING the OFFENDING AGENT
- * SEVERE CASES → CORTICOSTEROIDS



Slide 30: Silicosis TB risk, berylliosis, hypersensitivity pneumonitis treatment

🔄 **Recap: Exposures.** Hypersensitivity pneumonitis = organic antigen (birds, mouldy hay) → fever 4-8 h later, ground glass → avoid ± steroids. Pneumoconiosis = inorganic dust → macrophages → fibrosis. Asbestosis: lower lobes, pleural plaques, cancer and mesothelioma. Silicosis: upper lobes, eggshell nodes, TB (yearly TST). Byssinosis: cotton.

3. High-Yield Comparison Tables

Table A: Obstructive vs restrictive PFT [*obstructive column from the COPD/asthma lectures*]

	Obstructive	Restrictive
FEV1/FVC	↓ (< 70%)	Normal or ↑
FVC	Normal or ↓	↓
TLC, RV, FRC	↑ (air trapping)	↓
DLCO	↓ emphysema / normal asthma	↓ intrinsic / normal extrinsic

Table B: IPF vs COP vs sarcoidosis vs HP (*see the added diagram*)

Table C: Upper vs lower lobe

Upper lobe	Lower lobe
★ Silicosis , coal [<i>added</i>], sarcoidosis (reticular opacities in Case 2), HP (chronic) [<i>added</i>], TB	★ Asbestosis , IPF (basal subpleural)
<i>Mnemonic [added]: "SCHoT on top"</i> (Silicosis, Coal, HP, TB); "Asbestos sinks to the bases" .	

Table D: Steroids: yes or no?

Steroids help	Steroids NOT recommended
COP, radiation pneumonitis, sarcoidosis, HP (severe acute), berylliosis	★ IPF (immunosuppression worsens outcome) · asbestosis/silicosis (no specific therapy)

Table E: Sarcoid stages (*see §2.11*)

4. Rules, Exceptions, Patterns

- **Restriction** = FEV1/FVC normal or high + all volumes low.
- **Low DLCO** = parenchymal; normal DLCO suggests extrinsic [*added*].
- **Velcro** (fine, inspiratory, basal) crackles = fibrosis.
- **IPF**: older, slow, afebrile, basal subpleural honeycombing, UIP. Antifibrotics, not steroids.
- **COP**: acute, pneumonia-like, patchy, steroid-responsive.
- **Sarcoidosis**: non-caseating granulomas; caseating granulomas suggest TB [*added*].
- Sarcoid hypercalcaemia comes from calcitriol made by macrophages.
- Lupus pernio is pathognomonic; erythema nodosum is the commonest skin sign.
- **HP** = organic; pneumoconiosis = inorganic.
- **Asbestosis** → lower lobes, pleural plaques, cancer (adenocarcinoma) and mesothelioma; smoking multiplies risk.
- **Silicosis** → upper lobes, eggshell calcification, TB (annual TST).
- **Fever** is a feature of HP and sarcoidosis, not IPF.

5. Clinical Correlations

- **RA patient on methotrexate with dry cough, clubbing, basal crackles** → ILD (Case 1); consider **drug-induced** (methotrexate) vs **RA-ILD** [added].
- **Young woman with bilateral hilar nodes, hypercalcaemia, erythema nodosum and ankle pain** → sarcoidosis (Löfgren syndrome [added]) (Case 2).
- **Bilateral facial palsy + polyuria** → neurosarcoidosis (DI).
- **Pigeon keeper with recurrent fevers after cleaning lofts** → bird-breeder's lung.
- **Retired shipyard worker with pleural plaques and new effusion** → mesothelioma until proven otherwise.
- **Sandblaster with weight loss and cavitation** → silicotuberculosis.
- **Breast cancer patient 3 months after radiotherapy with cough and dyspnoea** → radiation pneumonitis → CT, steroids.
- **Drugs causing ILD (pleurisy list in the Pleura lecture):** amiodarone, bleomycin, methotrexate, nitrofurantoin [added].

6. Rapid Revision (Last-Day)

- Restrictive: ↓ volumes (TLC, VC, FRC, RV, VT); **FEV1/FVC normal or ↑**; ↓ **DLCO** if parenchymal.
- **Intrinsic:** ILD, sarcoid, pneumoconioses, ARDS, IPF, HP, GPA, radiation, drugs, alveolar filling. **Extrinsic:** pleural disease, kyphoscoliosis, ankylosing spondylitis, pregnancy, ascites, phrenic palsy, myasthenia.
- DPLD = progressive inflammation → fibrosis; repeated injury + aberrant healing → collagen → remodelling.
- Features: progressive dyspnoea, dry cough, **Velcro crackles**, fatigue, **clubbing**, fever (HP/sarcoid); late cyanosis, PH.
- CXR: reticular, reticulonodular, ground glass, **honeycombing**.
- Classification: known cause · **idiopathic interstitial pneumonias (IPF, DIP, RB-ILD, AIP, COP, NSIP, LIP)** · granulomatous (sarcoid) · other (LAM, HX, eosinophilic).
- **IPF:** most common ILD; > 50 y; slow; no fever; **UIP**; **basal subpleural honeycombing + traction bronchiectasis**; poor prognosis; **pirfenidone (TGF-β), nintedanib (TKI), transplant referral at diagnosis, no immunosuppression**.
- **COP:** days-weeks, mimics CAP, **patchy opacities, normal volumes**, 50% idiopathic, **steroids**.
- **AIP (Hamman-Rich):** rapid respiratory failure. **NSIP:** CTD (scleroderma), HIV, HP.
- **Radiation pneumonitis:** 5-15%; acute 1-6 months, chronic 1-2 years; **CT**; **steroids**.
- **Sarcoid:** non-caseating granulomas; AA women < 40; ⅔ remit; **bilateral hilar nodes, ↑ Ca²⁺ (calcitriol), ↑ ACE, erythema nodosum, lupus pernio**, anterior uveitis, facial palsy, DI, nephrocalcinosis; diagnosis of exclusion + biopsy; stages 0-4 (stage 1 remits 60-80%); **steroids ± DMARD**.
- **HP:** organic antigens; acute fever **4-8 h** after; ground glass; farmer's, bird-breeder's, air-conditioner, bagassosis, mushroom; **avoid ± steroids**.
- **Pneumoconiosis:** dust → macrophages → fibrosis. **Asbestosis** (lower lobes, > 15-20 y, plaques, cancer + mesothelioma, adenocarcinoma commonest). **Silicosis** (upper lobes, nodular, **eggshell**, **TB → yearly TST**). **Byssinosis** (cotton). **Berylliosis** → steroids.

7. Exam-Style Questions

Q1 (MCQ, lecture Case 1, Slide 31). 55 ♀ with **rheumatoid arthritis on methotrexate**, ex-smoker; 8 months of **dry cough and exertional dyspnoea**; ★ **clubbing**, ★ **bibasilar inspiratory crackles**; HRCT shown (reticulation/honeycombing); ★ **↓ DLCO**. Diagnosis? **f. Interstitial lung disease** **Answer: f.** [reasoning added; no key on the slide] Dry cough + progressive dyspnoea + basal Velcro crackles + clubbing + low DLCO + fibrotic HRCT = **ILD** (RA-associated and/or methotrexate-induced). **Caplan syndrome** = RA + pneumoconiosis with **nodules**, and she has no dust exposure (receptionist). Emphysema doesn't cause basal crackles or clubbing; effusion and oedema don't fit the HRCT.

Q2 (MCQ, lecture Case 2, Slide 32). 35 ♀, 6 months of dry cough and exertional dyspnoea, **bilateral ankle pain, non-tender axillary lymphadenopathy**, ★ **Ca²⁺ 11.7**, CXR ★ **bilateral hilar enlargement + upper-zone reticular opacities**. Underlying mechanism? **b. Granulomatous inflammation** **Answer: b. Sarcoidosis** (stage 2): non-caseating granulomas; hypercalcaemia from granuloma calcitriol; arthralgia/ankle pain. Necrotising inflammation suggests TB or GPA; neoplastic transformation suggests lymphoma (possible differential, but hypercalcaemia + hilar nodes + young woman fits sarcoid better).

Q3 (MCQ). A 68-year-old man has slowly progressive dyspnoea, Velcro crackles and basal subpleural honeycombing on HRCT. Which treatment is **inappropriate**? a) Nintedanib b) Pirfenidone c) Transplant referral **d) High-dose prednisone + azathioprine** **Answer: d.** Immunosuppression worsens outcomes in IPF.

Q4 (MCQ). A sandblaster should be screened annually for which condition? a) Mesothelioma **b) Tuberculosis** c) Sarcoidosis d) Byssinosis **Answer: b.** Silica impairs phagolysosome formation → TB (yearly TST).

Q5 (Short answer). Give three differences between asbestosis and silicosis. **Answer: Lobe:** asbestosis **lower**, silicosis **upper**. **Imaging:** asbestosis **pleural plaques**, silicosis **eggshell hilar calcification/nodules**. **Complication:** asbestosis → **bronchogenic carcinoma + mesothelioma**; silicosis → **TB**. (Exposure: shipbuilding/insulation vs mining/sandblasting/glass.)

8. Exam Pitfalls & Traps

1. **Restrictive FEV1/FVC is low.** No: **normal or ↑** (both FEV1 and FVC fall).
2. **Steroids for IPF.** Contraindicated; use **antifibrotics** and refer for transplant.
3. **IPF has fever.** No; fever points to **HP or sarcoid** (or COP).
4. **Sarcoid granulomas are caseating.** They're **non-caseating**.
5. **Sarcoid hypercalcaemia is from PTHrP.** No: **calcitriol** from granuloma macrophages (PTHrP = squamous cell lung cancer).
6. **HP vs pneumoconiosis: organic** antigens vs **inorganic** dust.
7. **Upper vs lower:** asbestosis lower; silicosis upper.
8. **Asbestos → squamous cancer.** The lecture says **adenocarcinoma** is the commonest; plus mesothelioma.
9. **COP = infection; give antibiotics only.** It mimics pneumonia but is **steroid-responsive**.
10. **Caplan syndrome** = RA + **pneumoconiosis** with lung nodules, not ordinary RA-ILD.
11. ⚠ **Lecture title vs file:** the file is called "Restrictive lung disease"; the slides are titled **Diffuse Parenchymal Lung Diseases**. Same content.
12. ⚠ **"Misc. DPLDs"** is in the subtopics but only LAM/HX/eosinophilic are named on the classification slide.